QDA Data — Multivariate Representation with Ellipses

Introduction

This analysis is designed for **Quantitative Descriptive Analysis (QDA)** data, that is, when several subjects assess several stimuli using a fixed list of sensory attributes. It extends the univariate characterisation of the sensory space by providing a multivariate representation of the stimuli and their variability. It addresses two complementary questions: **which stimuli are sensorially similar** and **is this proximity stable despite variability between subjects**? Its central result is a map of the stimuli enhanced with **ellipses**, which show not only the position of the stimuli but also the robustness of that position under panel resampling.

Why not use a standard alternative?

A PCA applied directly to the raw data may mix two sources of variation: the **stimulus** effect and the **subject** effect. In sensory analysis, however, some subjects systematically rate higher or lower than others, or use the scale more or less widely. This analysis takes that subject/stimulus structure into account in order to produce a representation better suited to QDA data.

Understanding what is happening "under the bonnet"

Selecting sensory attributes

Before constructing the map, the function identifies the attributes that discriminate sufficiently between stimuli. This selection depends on the value of **p-value of the Stimulus effect (%)**. The stricter the threshold, the more streamlined the map will be; the broader it is, the more exhaustive the representation will be, though sometimes more crowded. The default value is **20%**, which makes it a good starting point for exploration.

Correcting for the subject effect

A simple way to understand the next step is to say that the analysis reconstructs **mean sensory profiles by stimulus**, while taking differences between subjects into account. Two options are involved here:

- **Center by subject** corrects differences in mean level between subjects;
- **Scale by subject** corrects differences in the amplitude of scale use.

The aim is to prevent a subject who is very generous, very severe, or very "spread out" in their responses from artificially distorting the stimulus map.

Principal component analysis

A **PCA** is then performed on the resulting table of sensory profiles. It produces the stimulus map, the sensory attribute map, the eigenvalue table, and the automatic description of the dimensions. It is this factor space that then serves as the reference for projecting the virtual panels.

Virtual panels and supplementary individuals

The key feature of the module is the use of **resampling**. In PCA, it is possible to distinguish between **active individuals** — those used to construct the axes — and **supplementary individuals** — those projected afterwards without altering the axes. This is precisely the mechanism that makes it possible to construct the ellipses.

At each simulation, a subset of subjects is drawn with replacement from the original panel, new mean stimulus profiles are recalculated, and then projected into the factor space as supplementary individuals. The dispersion of these projections around each stimulus is then summarised by an ellipse.

These ellipses therefore describe the variability of the **mean position of the stimuli when the composition of the panel changes**, and not the mere raw dispersion of individual ratings.

> **Note:** the calculation may take a few seconds, as it relies on a large number of virtual panels. During this calculation, it is better to avoid changing other options so as not to restart the procedure unnecessarily.

Comparing stimuli

The function also calculates a comparison between pairs of stimuli, presented as a matrix of **p-values**. This output complements the visual reading of the ellipses by adding a numerical measure of the difference between stimuli.

Automatic description of the dimensions

Finally, the function provides an automatic description of the axes, filtered according to the value of **Significance threshold (%)**. It helps to identify which attributes are most associated with each dimension and to give the axes a *sensory* meaning.

Expected data structure

The data must contain:

- a column identifying the **stimuli**;
- a column identifying the **subjects**;
- several numeric columns corresponding to the **sensory attributes**.

Each row corresponds to the evaluation of a stimulus by a subject. If a subject evaluates the same stimulus several times (repetitions or sessions), each evaluation must appear on a separate row.

Before running the analysis in jamovi, check in the **Data** tab that:

- the variable intended for **Stimulus Effect** is indeed categorical;
- the variable intended for **Subject Effect** is indeed categorical;
- all variables intended for **Sensory Attributes** are numeric.

Guide to the interface

The analysis appears in the menu **SEDA > Fixed List of Attributes > Representation of the Stimulus Space**. The interface includes three variable drop zones, built-in help, and then three collapsible menus: **Graphic Options**, **Resampling Options**, and **Numerical Indicators**.

Variables to enter

Stimulus Effect

Role: identifies the stimuli to be represented on the map. **Impact on the calculation:** each level of this variable becomes a mean point and, where appropriate, an ellipse on the map. **Practical advice:** use clear labels, as they will appear directly in the graphs.

Subject Effect

Role: identifies the subjects. **Impact on the calculation:** this variable is essential for correcting the subject effect and for generating the virtual panels used in the bootstrap. **Practical advice:** check that the same subject keeps exactly the same identifier throughout the file.

Sensory Attributes

Role: groups together the sensory attributes used to construct the factor space. **Impact on the calculation:** only numeric attributes are accepted; the number of dimensions that can be calculated depends on the number of retained attributes. **Practical advice:** start by including all relevant sensory attributes, then refine with **p-value of the Stimulus effect (%)**.

Read me before running

Role: displays built-in help text in the results. **Impact on the calculation:** none. **Practical advice:** leave this option ticked for the first few uses. The default value is **enabled**.

Standardization of the Sensory Attributes: Scale to unit variance

Role: standardises the sensory attributes when performing the PCA. **Impact on the calculation:** when ticked, each attribute contributes equally to the construction of the axes; when unticked, the most dispersed attributes have a greater influence on the construction of the axes. **Practical advice:** leave this option ticked in most analyses. It is **enabled by default**.

Selection of the Sensory Attributes: p-value of the Stimulus effect (%)

Role: sets the threshold for selecting attributes according to their link with the stimulus effect. **Impact on the calculation:** a low threshold retains few attributes and produces a clearer map; a higher threshold retains more attributes and gives a broader view. **Practical advice:** the default value is **20%**, a good starting point for exploration. For a stricter version, begin at **5%**.

Graphic Options

Components to Plot — X-axis and Y-axis

Role: choose the dimensions displayed on the x-axis and y-axis. **Impact on the calculation:** they do not change the PCA itself, but they alter the displayed factor plane. The selected dimensions must exist in the analysis, otherwise the function generates an error. **Practical advice:** start with **X-axis = 1** and **Y-axis = 2**, the default values.

Individual variability around stimuli

Role: adds a graph showing the dispersion of individual evaluations around each stimulus. **Impact on the calculation:** it does not add a new analysis, but an additional graphical output useful for assessing consensus between subjects. **Practical advice:** activate this option if you wish to distinguish a consensual stimulus from one perceived more variably. The default value is **disabled**.

Variability around sensory attributes

Role: adds a graph showing the variability of the sensory attributes under resampling. **Impact on the calculation:** it provides a graphical output useful for assessing the stability of the interpretation of the attributes. **Practical advice:** activate this option when you want to comment on the robustness of the axes. The default value is **disabled**.

Resampling Options

Number of subjects

Role: defines the number of subjects drawn in each virtual panel. **Impact on the calculation:** the closer this value is to the real size of the panel, the more similar the simulated panels are to it. **Practical advice:** use the real panel size as a starting point. The default value is **20**.

Number of panels

Role: sets the total number of simulated virtual panels. **Impact on the calculation:** the larger this number, the more stable the ellipses are from one run to another, but the calculation time increases. **Practical advice: 300** is a good compromise to begin with; for a final version, it may be useful to increase this number to **500**. The default value is **300**.

Threshold (%)

Role: defines the level used to draw the ellipses. **Impact on the calculation:** a low threshold produces larger ellipses corresponding to a higher confidence level. **Practical advice:** leave **5%** for a first attempt. This is the default value.

Center by subject

Role: centres responses subject by subject. **Impact on the calculation:** it corrects differences in mean level between subjects and often improves profile comparison. **Practical advice:** leave this option ticked in the vast majority of analyses. It is **enabled by default**.

Scale by subject

Role: standardises responses subject by subject. **Impact on the calculation:** it corrects differences in amplitude between subjects, but modifies the initial structure more strongly than centring alone. **Practical advice:** leave this option unticked at first. It is **disabled by default**.

Numerical Indicators

Significance threshold (%)

Role: defines the significance threshold for the automatic description of the dimensions. **Impact on the calculation:** the stricter the threshold, the more selective the automatic description. **Practical advice:** the default value, **5%**, is well suited to a first reading.

Number of dimensions

Role: defines the number of dimensions to be calculated and described. **Impact on the calculation:** this value is limited by the dimensions that can actually be calculated by the PCA. **Practical advice:** the default value, **2**, is appropriate for an initial reading.

Step-by-step procedure

1. Open your dataset in jamovi.
2. In the **Data** tab, check that the stimulus variable is categorical, the subject variable is categorical, and the sensory attributes are numeric.
3. Open **SEDA > Fixed List of Attributes > Representation of the Stimulus Space**.
4. Place the stimulus variable in **Stimulus Effect**.
5. Place the subject variable in **Subject Effect**.
6. Place all sensory attributes in **Sensory Attributes**.
7. Leave **Scale to unit variance** ticked.
8. Set **p-value of the Stimulus effect (%)** to **20%** for initial exploration, or to **5%** for a stricter selection.
9. Keep **X-axis = 1** and **Y-axis = 2**.
10. Leave **Center by subject** ticked and **Scale by subject** unticked to begin with.
11. Set **Number of subjects** to the panel size or a similar value, leave **Number of panels** at **300**, and **Threshold (%)** at **5**.
12. Activate **Individual variability around stimuli** and/or **Variability around sensory attributes** as needed.
13. In **Numerical Indicators**, keep the default values for a first attempt.

14. First read **Representation of the Stimuli with Ellipses**, then **Eigenvalue Decomposition**, then **Automatic Description of the Dimensions**.
15. Then go back to the settings to test the stability of your conclusions.

Interpreting the results

Representation of the Stimuli with Ellipses

This is the central graph of the module. Each stimulus is represented by a mean point in the factor space, surrounded by an ellipse. The axes come from the PCA, and the percentage displayed on each axis indicates the share of explained variance.

Element	Meaning
Two nearby stimuli	Similar sensory profiles
Strongly overlapping ellipses	Difference is not very significant (with respect to a change in panel)
Small ellipse	Stable position across simulated panels
Large ellipse	More uncertain position
Clearly separated ellipses	More significant difference

> **Point of attention:** the ellipses describe the variability of the mean position of the stimuli under panel resampling. They do not directly represent the raw dispersion of individual ratings.

Representation of the Stimuli

This map shows the relative positions of the stimuli without emphasising the ellipses. It is useful for quickly identifying groupings and major oppositions between stimuli.

Representation of the Sensory Attributes

This map shows the sensory attributes in the factor space. It can be read according to three simple cues:

- **direction**: towards which stimuli the attribute "pulls" the structure;
- **length**: the longer an attribute is, the better it is represented on the plane;
- **angle between two attributes**: a small angle = positive correlation; an angle close to 180° = opposition; an angle close to 90° = near-independence.

This is the map that gives the axes a sensory meaning.

Individual variability around stimuli *(available only if the option is activated)*

This output shows the dispersion of individual evaluations around the mean point of each stimulus. A compact cloud indicates strong consensus; a dispersed cloud indicates more heterogeneous perception between subjects.

Variability around sensory attributes *(available only if the option is activated)*

This output shows the dispersion of the position of the sensory attributes in the virtual panels. A compact cloud means that the attribute is stable; a dispersed cloud indicates that its interpretation is more fragile and depends more heavily on the composition of the panel.

Eigenvalue Decomposition

This table gives, for each dimension, the eigenvalue, the percentage of explained variance, and the cumulative percentage. If the first two dimensions explain a large share of the variance, the 2D map summarises the data well. Otherwise, it is prudent to explore other planes, for example Dim 1 / Dim 3.

Automatic Description of the Dimensions

This textual output identifies the attributes most closely linked to each dimension, filtered by **Significance threshold (%)**. It helps to give the axes sensory names and complements the attribute map well.

Product-by-product comparison table *(available only if Number of subjects > 1)*

This matrix of p-values compares each pair of stimuli in the factor space. A low p-value suggests a clearer difference between two stimuli; a high p-value suggests a more plausible proximity. This table complements the visual reading of the ellipses, especially when their overlap is difficult to interpret by eye.

Worked example

The **sensochoc** dataset describes 6 chocolates evaluated by 29 subjects on 14 sensory attributes. To load this dataset, open jamovi, click the icon at the top left (three horizontal lines), then go to **Open** → **Data Library** → **SEDA** to find the **sensochoc** dataset.

Recommended settings

Option	Value
Stimulus Effect	Product
Subject Effect	Panelist
Sensory Attributes	all sensory attributes
Scale to unit variance	Ticked
p-value of the Stimulus effect (%)	20
X-axis / Y-axis	1 / 2
Number of subjects	29
Number of panels	300
Threshold (%)	5
Center by subject	Ticked
Scale by subject	Unticked
Significance threshold (%)	5
Number of dimensions	2

Reading the Representation of the Stimuli with Ellipses map

The map shows the 6 chocolates on the Dim. 1 / Dim. 2 plane. The first dimension alone accounts for **88.79%** of the total variance — so this 2D plane is a very faithful representation of the data. The second dimension explains only an additional 7.58%.

On the horizontal axis (Dim. 1), **choc1** and **choc3** occupy opposite extremes: **choc1** on the left (bitter, cocoa-like side), **choc3** on the right (milky, sweet side). These are the two chocolates whose profiles are most clearly opposed, which is consistent with the results of the previous chapter. **choc2**, **choc5**, and **choc6** are grouped in the centre, near the origin. **choc4** lies slightly above the horizontal axis, between choc1 and the central group.

The ellipses indicate how robust these positions are. The ellipses for **choc2**, **choc5**, and **choc6** overlap widely. In other words, if the composition of the panel changed, their order could be reversed. By contrast, **choc1** and **choc3** have ellipses clearly separated from everything else: their position is stable, and their difference from the other chocolates can be considered reliable.

Reading the Hotelling Test table

The *p-value* table quantitatively confirms what the ellipses show visually. The pairs **choc2 / choc5** (p = 0.76) and **choc5 / choc6** (p = 0.16) have high p-values: one cannot conclude that there is a significant difference between these chocolates in the sensory space. The pair

choc2 / choc6 is borderline (p = 0.04). By contrast, all comparisons involving **choc1** or **choc3** give p-values of 0.00: these two chocolates differ significantly from all the others.

Reading the Representation of the Sensory Attributes map

The 14 attributes are distributed very clearly around the horizontal axis. On the right are grouped *MilkF*, *Sweetness*, *Caramel*, *Vanilla*, *MilkA*, and *Melting* — this is the profile of **choc3**. On the left are grouped *CocoaF*, *CocoaA*, *Bitterness*, *Astringency*, *Acidity*, *Crunchy*, and *Granular* — this is the profile of **choc1**. The horizontal axis therefore clearly corresponds to an opposition between **sweet milk chocolate / bitter dark chocolate**.

The length of the arrows also indicates the quality of representation of each attribute on this plane. *Crunchy* has a shorter arrow and is oriented downwards, which reflects part of its variability on the second dimension — confirming that this attribute is less stable in the subjects' perception than *CocoaF* or *MilkF*, whose arrows almost reach the edge of the circle.

Reading the Automatic Description of the Dimensions table

The automatic description of the dimensions confirms these readings. On Dim. 1, the attributes most positively correlated are *MilkF* (correlation 0.86), *Sweetness* (0.66), and *Caramel* (0.65); the most negatively correlated are *CocoaF* (-0.79) and *Bitterness* (-0.70). On Dim. 2, *Sticky* (0.59), *Acidity* (0.58), and *Astringency* (0.57) stand out positively, while *Melting* (-0.15) is slightly negative.

Synthetic interpretation

The sensory space of the chocolates is largely structured by one strong opposition, carried by the first dimension: on one side a milky, sweet, and melting profile; on the other a bitter, cocoa-like, and more textural profile. The stimulus map and the attribute map converge on the same overall reading. The ellipses additionally show that not all stimuli are differentiated with the same robustness. Finally, the Hotelling test confirms that some proximities observed on the map do not correspond to clearly established differences.

> **What this example shows:** the analysis does not simply show where the stimuli are located in the sensory space, but also how robustly they can be distinguished.

Practical advice

- Always start with **X-axis = 1** and **Y-axis = 2**.
- Leave **Scale to unit variance** ticked in most cases.
- Use **Center by subject** almost systematically.
- Only activate **Scale by subject** if you have good reason to think that subjects use the scale with very different amplitudes.
- Start with **300** simulated panels; increase to **500 or more** for a more stable final version.
- Set **p-value of the Stimulus effect (%)** between **5%** and **20%** depending on whether you want a streamlined or a more exhaustive map.
- Never interpret the stimulus map without also looking at the attribute map.
- Do not conclude too quickly that there is a difference when the ellipses overlap strongly.
- Consult **Product-by-product comparison table** to complement the visual reading when the overlap is ambiguous.
- If the first two dimensions summarise little variance, explore other factor planes.

In summary

This analysis produces a multivariate representation of stimuli suited to QDA data, taking the subject effect into account when constructing sensory profiles. It enriches the classical factor map with ellipses obtained by resampling, making it possible to assess the robustness of the observed positions. It therefore shows not only **where the stimuli are located**, but also **with what degree of confidence they can be placed there**.